Debangan Dey, Ph.D.

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Research Interests

Summary — A statistician focused on developing methods to analyze intensive multilevel multimodal longitudinal data and big geospatial data. Applications focus on Digital Health Technologies (DHTs) and Environmental Sciences.

Areas — Functional data analysis, longitudinal data, spatial statistics, stochastic processes, semiparametric statistics.

Education

2017 – 2022 — **Ph.D., Johns Hopkins Bloomberg School of Public Health** in Biostatistics.
Thesis title: *Topics in Modeling of Multivariate Mixed Data Types and Highly Multivariate Spatial Data*

2015 – 2017 — **Masters, Indian Statistical Institute** in Statistics.
Thesis title: *On Investigation of Genotypic Association with Longitudinal Observations of Traits*.

2012 – 2015 — **Bachelors, Indian Statistical Institute** in Statistics.

Professional Experience

07/2025 – present — **Assistant Professor**, Department of Statistics, Texas A&M University.

05/2022 – 05/2025 — **Postdoctoral Fellow**, National Institute of Mental Health (NIMH), NIH. Lead analyst for the Motor Activity Research Consortium for Health (mMARCH). Advisor: Dr. Kathleen R. Merikangas.

05/2021 – 08/2021 — **Applied Scientist Intern**, Amazon. Developed heterogeneous treatment effect methods to reduce A/B testing trial length in the Amazon Search team.

2020 — **Analyst**, Baltimore City Health Department – Johns Hopkins Data Analysis Support Team for COVID-19 Response.

2015, 2016 — **Summer Intern**, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health.

Research Publications and Repositories

Journal Articles

1. **D. Dey**, H. A. Lateef, A. Leroux, V. Zipunnikov, and K. Merikangas, "Associations between daily outdoor temperature and subjective real-time ratings of emotional states and sleep in mood disorder subtypes," *Journal of Affective Disorders*, vol. 397, p. 120 918, 2026.

2. B. Lim, S. A. Bruce, **D. Dey**, A. MacMillan Uribe, J. Szeszulski, R. A. Seguin-Fowler, and C. D. Rethorst, "Interpretable machine learning for predicting glucose excursions in daily life using multimodal behavioral data," *Annals of Behavioral Medicine*, 2026, Accepted for publication.

3. M. Shin, E. Tonini, J. S. Carpenter, M. Varidel, A. Nichles, N. Zmicerevska, E. Phung, C. Janiszewski, M. Park, D. Froggatt, D. Hanlon, A. Chami, M. K. Chong, H. LaMonica, F. Iorfino, A. C. Burns, S. W. Cain, S. J. Kang, V. Zipunnikov, W. Guo, **D. Dey**, A. Leroux, K. R. Merikangas, E. M. Scott, I. B. Hickie, and J. J. Crouse, "Seasonal variation in mood and associated patterns of sleep, activity, circadian rhythms, and light," *Journal of Biological Rhythms*, 2026, In press. DOI: 10.1177/07487304261496959.

4. **D. Dey**, S. Banerjee, M. A. Lindquist, and A. Datta, "Graph-constrained analysis for multivariate functional data," *Journal of Multivariate Analysis*, vol. 207, p. 105 428, 2025.

5. **D. Dey**, R. Ghosal, K. Merikangas, and V. Zipunnikov, "Functional principal component analysis for continuous non-Gaussian, truncated, and discrete functional data," *Statistics in Medicine*, vol. 43, no. 28, pp. 5431–5445, 2024.

6. S. J. Kang, A. Leroux, W. Guo, **D. Dey**, M.-P. F. Strippoli, J. Di, J. Vaucher, P. Marques-Vidal, P. Vollenweider, M. Preisig, K. R. Merikangas, and V. Zipunnikov, "Integrative modeling of accelerometry-derived sleep, physical activity, and circadian rhythm domains with current or remitted major depression," *JAMA Psychiatry*, vol. 81, no. 9, pp. 911–918, 2024.

7. T. M. Lateef, **D. Dey**, A. Leroux, L. Cui, M. Xiao, V. Zipunnikov, and K. R. Merikangas, "Association between electronic diary–rated sleep, mood, energy, and stress with incident headache in a community-based sample," *Neurology*, vol. 102, no. 4, e208102, 2024, **Article featured at CNN, Nat Geo, NIH Director's blog and others**.

8. K. R. Merikangas, S. J. Kang, F. Lamers, A. Leroux, **D. Dey**, M. M. Husky, M. Preisig, I. Hickie, and V. Zipunnikov, "Harmonizing real time multi-domain assessments of mood disorder spectrum: Motor Activity Research Consortium for Health (mMARCH)," *Bipolar Disorders*, vol. 26, no. Suppl 1, p. 34, 2024.

9. **D. Dey**, A. Datta, and S. Banerjee, "Modeling multivariate spatial dependencies using graphical models," *The New England Journal of Statistics in Data Science*, vol. 1, no. 2, pp. 283–295, 2023.

10. J. Glaus, S. J. Kang, W. Guo, F. Lamers, M.-P. F. Strippoli, A. Leroux, **D. Dey**, K. J. Plessen, J. Vaucher, P. Vollenweider, *et al.*, "Objectively assessed sleep and physical activity in depression subtypes and its mediating role in their association with cardiovascular risk factors," *Journal of Psychiatric Research*, vol. 163, pp. 325–336, 2023.

11. E. K. Stapp, V. Zipunnikov, A. Leroux, L. Cui, M. M. Husky, **D. Dey**, and K. R. Merikangas, "Specificity of affective dynamics of bipolar and major depressive disorder," *Brain and Behavior*, vol. 13, no. 9, e3134, 2023.

12. **D. Dey**, A. Datta, and S. Banerjee, "Graphical Gaussian process models for highly multivariate spatial data," *Biometrika*, vol. 109, no. 4, pp. 993–1014, 2022.

13. M. R. Kaufman, **D. Dey**, C. Crainiceanu, and M. Dredze, "#MeToo and Google inquiries into sexual violence: A hashtag campaign can sustain information seeking," *Journal of Interpersonal Violence*, vol. 36, no. 19-20, pp. 9857–9867, 2021.

14. **D. Dey** and V. Zipunnikov, "Discussion of "an epidemiological forecast model and software assessing interventions on the COVID-19 epidemic in China"," *Journal of Data Science*, vol. 18, no. 3, pp. 433–436, 2020.

15. S. Deb and **D. Dey**, "Spatial modeling of shot conversion in soccer to single out goalscoring ability," *Journal of Sports Analytics*, vol. 5, no. 4, pp. 281–297, 2019.

16. V. R. Varma, **D. Dey**, A. Leroux, J. Di, J. Urbanek, L. Xiao, and V. Zipunnikov, "Total volume of physical activity: TAC, TLAC or TAC (λ)," *Preventive Medicine*, vol. 106, pp. 233–235, 2018.

17. A. Spira, V. Zipunnikov, M. Wu, **D. Dey**, E. Simonsick, L. Ferrucci, C. Davatzikos, and S. Resnick, "Association of circadian rest and activity rhythms with brain volumes in cognitively normal older adults," *Innovation in Aging*, vol. 1, no. suppl_1, pp. 866–866, 2017.

18. V. R. Varma, **D. Dey**, A. Leroux, J. Di, J. Urbanek, L. Xiao, and V. Zipunnikov, "Re-evaluating the effect of age on physical activity over the lifespan," *Preventive Medicine*, vol. 101, pp. 102–108, 2017, **Article featured at TIME, Washington Post, WSJ, BBC, WPYR and others**.

19. V. Zipunnikov, **D. Dey**, A. Leroux, J. Di, J. Urbanek, J. Schrack, and C. Crainiceanu, "Total physical activity and its circadian allocation are independent predictors of mortality," *Innovation in Aging*, vol. 1, no. Suppl 1, p. 1239, 2017.

Preprints and Under Revision

1. A. Burman, S. Choudhury, and **D. Dey**, *Doubly-unlinked regression for dependent data*, 2026. arXiv: 2603.19506 [math.ST]. URL: https://arxiv.org/abs/2603.19506.

2. **D. Dey**, *Do in-match hydration breaks alter match momentum? a within-match case-crossover analysis of the 2026 FIFA World Cup*, 2026. arXiv: 2607.19783 [stat.AP]. URL: https://arxiv.org/abs/2607.19783.

3. **D. Dey**, R. Ghosal, K. Merikangas, and V. Zipunnikov, *Multivariate functional principal component analysis for mixed-type mHealth data: An application to mood disorders*, 2026. arXiv: 2603.11385 [stat.ME]. URL: https://arxiv.org/abs/2603.11385.

4. **D. Dey**, A. Manna, and C. J. Geoga, *Bigraphical Matérn–Whittle (BMW) processes for fast inference of big multivariate spatial data on general domains*, 2026. arXiv: 2609.01950 [stat.ME].

5. **D. Dey** and V. Zipunnikov, *Regression and dimension reduction for multivariate mixed-type data via semiparametric Gaussian copula*, 2026. arXiv: 2205.06868 [stat.ME]. URL: https://arxiv.org/abs/2205.06868.

6. M. M. Husky, L. Hunt, A. Leroux, **D. Dey**, K. Conway, C. Vetter, V. Zipunnikov, R. Choudhari, M. Kass, M. Leyden, Y. Kotani, T. Lateef, G. Salum, A. Swaminathan, Y. Xiao, A. Klein, M. Milham, and K. R. Merikangas, *Real-time Ecological Assessment of the Context of mental and physical Health (REACH) on Curious*, Research Square [Preprint]. PMID: 41472691, 2025. URL: https://doi.org/10.21203/rs.3.rs-8171682/v1.

7. **D. Dey** and V. Zipunnikov, *Connecting population-level auc and latent scale-invariant R^2 via semiparametric Gaussian copula and rank correlations*, 2019. arXiv: 1910.14233 [stat.ME].

Packages and Repositories

1. **D. Dey**, *R package – M2FPCA: Multivariate functional principal component analysis for mixed-type mHealth data*, https://github.com/Ddey07/M2FPCA, 2026.

2. **D. Dey** and R. Ghosal, *R package – SGCTools: Analysis tools for multivariate mixed data and continuous/truncated/discrete functional data using Semiparametric Gaussian Copula (SGC)*, https://github.com/Ddey07/SGCTools, 2023.

3. **D. Dey** and A. Pita, *The Good, The Bad, and The Ugly of the Beautiful Game — Analyzing FIFA World Cup 2018 data*, https://ddey07.github.io/open-data/, 2018.

Books and Chapters

1. S. Deb and **D. Dey**, *Mathematical Techniques for Competitive Examinations*. Universities Press (India), 2023, ISBN: 978-93-93330-10-9.

Grants and Funding

Funded Support

2026–2027 **NIH Director's Challenge Innovation Award** — *Biologic Rhythms and Environmental Contexts in Human Health: Integrating Epidemiology and Circadian Science*. PI: K. Merikangas (NIMH). Total: $192,000 (FY2026) + $213,064 (FY2027). Role: Biostatistician Consultant (IPA, Texas A&M); leading data management and analytic strategies for multilevel circadian and metabolic data integration.

2026 Intergovernmental Personnel Act (IPA) Assignment with National Institute of Mental Health (NIMH), NIH. $35,147.54 (summer appointment). Role: Appointee. Organizing and conducting collaborative research on real-time ecological assessment of mental and physical health outcomes.

Talks and Presentations

Invited Talks

2026 "Doubly Unlinked Regression for Dependent Data." Invited session, International Indian Statistical Association (IISA) 2026 Conference, Banaras Hindu University, Varanasi, India. December 2026.

"Bigraphical Matérn–Whittle Processes for Fast Inference of Big Multivariate Spatial Data on General Domains." Statistics Seminar Series, Department of Mathematics and Statistics, University of Maryland, Baltimore County (UMBC). October 2026.

"Digital Phenotyping for Mixed-Type mHealth Data." Invited session, 33rd International Biometric Conference (IBC 2026), Seoul, South Korea. 2026.

2025 "Digital Phenotyping for Mixed-Type mHealth Data." Interdisciplinary Statistical Research Unit (ISRU), Indian Statistical Institute, Kolkata, India. December 2025.

"Innovations in Multivariate Stochastic Processes: Bridging Environment and Digital Health." STATCafe Graduate Seminar Series, Department of Statistics, Texas A&M University. September 2025.

"Innovations in Multivariate Stochastic Processes: Bridging Environment and Digital Health." Precise Advanced Technologies and Health Systems for Underserved Populations (PATHS-UP), Texas A&M University. September 2025.

"Graphical Methods for Multivariate Spatial Data." Interdisciplinary Hydrology Seminar Series, Departments of Biological and Agricultural Engineering and Water Management and Hydrologic Sciences, Texas A&M University. September 2025.

Contributed Talks

"Multivariate Principal Component Analysis for Mixed-Type Functional Data with Application to mHealth in Mood Disorders." 19th International Conference on Computational and Methodological Statistics (CMStatistics 2025), Birkbeck, University of London, UK. December 2025.

2024 "Multivariate Principal Component Analysis for Mixed-Type Functional Data with Application to mHealth." Joint Statistical Meetings (JSM 2024), Portland, OR.

2023 "Covariance Estimation and Functional Principal Component Analysis for Mixed-Type Functional Data." Joint Statistical Meetings (JSM 2023), Toronto, Canada.

2021 "Semiparametric Gaussian Copula Regression Modelling of Mixed Data Types." ENAR Spring Meeting 2021.

"Using Mobile Technologies to Investigate Impaired Sleep, Mood, and Energy as Real-Time Triggers of Migraine." International Conference on Ambulatory Monitoring of Physical Activity and Movement (ICAMPAM 2021).

Talks and Presentations (continued)

- "Modelling of Mixed-Type Intensive Longitudinal Data via Semiparametric Gaussian Copula and Its Application to Real-Time Mobile Monitoring of Daily Health Behaviors." Mobile Apps and Sensors in Surveys (MASS 2021).
- "Graphical Gaussian Process Models for Highly Multivariate Spatial Data." Joint Statistical Meetings (JSM 2021). *SBSS Student Paper Competition Winner.*

2020
- "Connecting Population-Level AUC and Latent R^2 via Semiparametric Gaussian Copula and Rank Correlations." Joint Statistical Meetings (JSM 2020). *JSM Student Travel Award Winner (Survey Research Methods Section).*
- "Connecting Population-Level AUC and Latent R^2 via Semiparametric Gaussian Copula." ENAR Spring Meeting 2020.

2019
- "Joint Modeling of Binary and Continuous Measurements in Large Health Surveys and Its Application to Network Analysis, Frailty, and Mortality in NHANES 1999–2010." ENAR Spring Meeting 2019.

Poster Presentations

2022
- "Influences of Mood, Energy and Sleep on Incident Headache Assessed with Prospective Real-Time Electronic Assessments." 24th Annual NIMH IRP Scientific Training Day.

Session Organizing

2025
- Invited session proposal accepted: "Data-driven Advances in Mental Health Statistics: Novel Methods in mHealth and Neuroimaging." 33rd International Biometric Conference (IBC 2026), Seoul, South Korea.
- Topic-contributed session proposal accepted: "Data-driven Advances in Mental Health Statistics: Novel Methods in mHealth and Neuroimaging." Joint Statistical Meetings (JSM 2026).

2023
- Organized invited session: "Recent Developments in Methods for Digital Brain Health Data." Joint Statistical Meetings (JSM 2023), Toronto, Canada.

Teaching

Courses Taught

Spring 2026
- **STAT 632**: Statistical Methodology II — Bayesian Modeling and Inference, Texas A&M University.

Fall 2025
- **STAT 641**: The Methods of Statistics I — Sections 600 (in-person) and 700 (web), Texas A&M University.

Teaching Assistantships

2019–2022
- Statistical Methods in Public Health I–IV (PH.140.621–624), Johns Hopkins Bloomberg School of Public Health.

2018–2019
- Probability Theory I–IV (PH.140.721–724), Johns Hopkins Bloomberg School of Public Health.

Other Teaching

2015–2017
- Instructor, Ramanujan School of Mathematics. Taught advanced high school mathematics for two years.

Mentoring

PhD Student Mentoring

2026– Gayun Kwon, Statistics PhD student, Texas A&M University.

Swapnonil Mondal, Statistics PhD student, Texas A&M University.

Other Mentoring

2022–2024 Ananya Swaminathan, PostBac, National Institute of Mental Health. Statistical components of research projects.

Rene Choudhari, PostBac, National Institute of Mental Health. Statistical components of research projects.

2020–2022 Final-term capstone project mentor for diverse cohorts of international students (25 countries) from non-statistical backgrounds, MPH program at Johns Hopkins Bloomberg School of Public Health.

Awards and Honors

2022 **Inducted into the Alpha chapter of Delta Omega**, national honorary society in public health.

2021 **Winner, Student Paper Competition**, *Section on Bayesian Statistical Science*, American Statistical Association (JSM 2021).

Selected among top six presenters, Opta Pro Forum, 2021, one of the biggest soccer analytics conferences attended by professionals from over 80 clubs and federations worldwide.

2020 **Winner, Student Paper Competition** (JSM Student Travel Award), *Survey Research Methods Section*, American Statistical Association (JSM 2020).

2018 **Top five finalist**, US Soccer Hackathon 2018, Chicago, USA, July 14–15, 2018.

2012 **INSPIRE scholarship**, Department of Science and Technology, Govt. of India.

2011, 2012 **Awardee**, Regional Math Olympiad, India.

2011 **Top 5%**, UNESCO Science Olympiad.

Professional Activities and Service

Affiliations and Research Groups

2025– Affiliated Member, Center for Biological Clocks Research (CBCR), Texas A&M University.

Active member, Space Time Research Group, Department of Statistics, Texas A&M University.

University Service

Member, Undergraduate Program Committee, Department of Statistics, Texas A&M University.

2025 Judge, poster competition at the 8th Annual Symposium on Cancer Research: Basic Science to Bioinformatics, Rudder Tower, Texas A&M University (in collaboration with the Center for Statistical Bioinformatics).

Professional Activities and Service (continued)

Editorial and Peer Review Activities

2020– Peer reviewer for: *Biometrics*, *Biostatistics*, *Statistics in Medicine*, *Journal of Machine Learning Research*, *Journal of Computational and Graphical Statistics*, *Annals of Applied Statistics*, *Econometrics and Statistics*, *TEST*, *Environmetrics*, *Scientific Reports*, *Sensors*, *Digital Health*, *International Journal of Environmental Research and Public Health*, *Mental Health and Physical Activity*, *Journal of Interpersonal Violence*, *Journal of Sports Analytics*.

Technical Skills

Programming R, Python, HTML, LaTeX

Languages English, Bengali, Hindi.

References

Dr. Vadim Zipunnikov, *PhD advisor*
Professor
Johns Hopkins Bloomberg School of Public Health,
Dept. of Biostatistics
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Dr. Abhirup Datta, *PhD advisor*
Associate Professor
Johns Hopkins Bloomberg School of Public Health,
Dept. of Biostatistics
Email: abhidatta@jhu.edu

Dr. Kathleen Merikangas, *Postdoc advisor*
Chief of the Genetic Epidemiology Research Branch
National Institute of Mental Health,
NIH
Email: merikank@mail.nih.gov

Dr. Sudipto Banerjee, *Collaborator*
Professor
University of California Los Angeles,
Fielding School of Public Health,
Dept. of Biostatistics
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